Student Performance Analysis Report

1. Introduction

Student performance analysis is an important task in education to understand the factors that influence academic success. By analyzing student data, institutions can identify patterns and improve teaching strategies.
This project focuses on analyzing student grades using **Python, Pandas, Matplotlib, and Seaborn**. The goal is to explore relationships between study habits, gender, and academic performance.

2. Objective

The main objectives of this project are:

- To analyze student grades (G1, G2, G3)
- To identify pass and fail distribution
- To study the impact of study time on performance
- To compare performance based on gender
- To find correlations between different factors

3. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

4. Dataset Description

The dataset contains student academic records including:

Column	Description
G1	First Period grade
G2	Second Period grade
G3	Final grade
Sex	Gender of student
Study time	Weekly study time
Absences	Number of absences

5. Data Exploration

Basic dataset exploration was performed using:
df.head()
df.shape
df.describe()
df.columns
This helped understand:

- Dataset size
- Statistical summary
- Column structure

6. Data Cleaning

Missing values were checked using:
df.isnull().sum()
No major missing values were found, so the dataset was ready for analysis.

7. Data Analysis and Visualization

7.1. Average Student Grades

```
avg_scores = df[['G1','G2','G3']].mean()
avg_scores.plot(kind='bar')
```

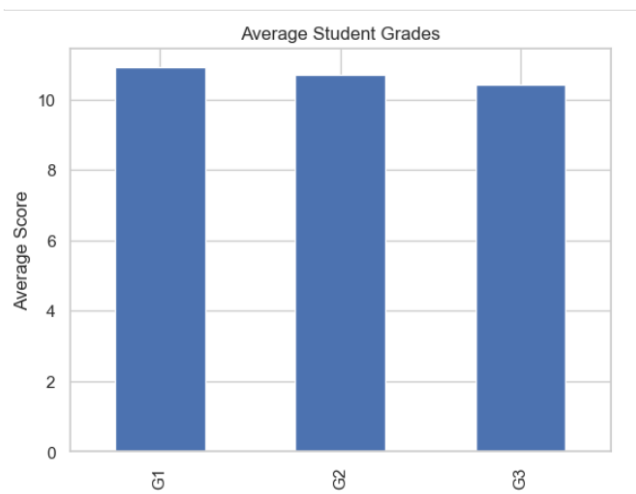

Figure 1: Average Student Grades

Interpretation:

The bar chart shows the average performance of students across three exams. It helps understand overall academic trends.

7.2. Pass vs Fail Distribution

```
df["Pass"] = df["G3"] >= 10
```

```
pass_count = df["Pass"].value_counts()
pass_count.plot(kind="pie", autopct="%1.1f%%")
```

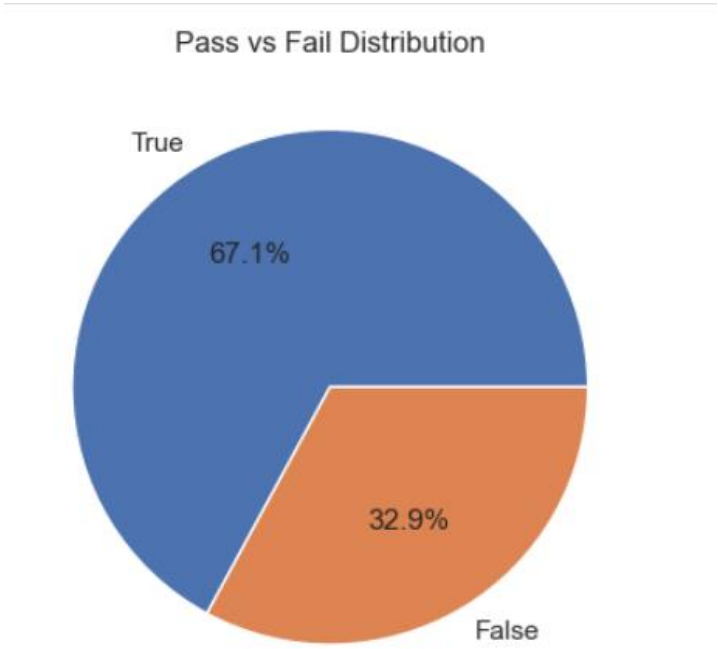

Fig2: Pass vs Fail Distribution

Interpretation:
This chart shows the percentage of students who passed and failed. It gives a clear understanding of overall success rate.
Performance by Gender

7.3. Performance by Gender

```
gender_scores = df.groupby("sex")[['G1','G2','G3']].mean()
gender_scores.plot(kind="bar")
```

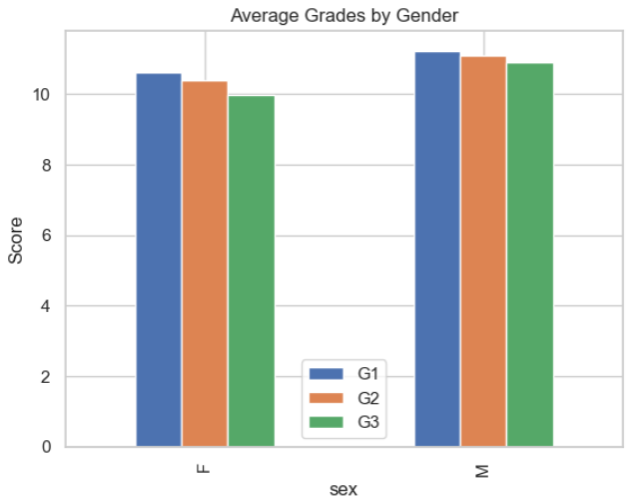

Fig3: Average grades by gender

Interpretation:
This analysis compares academic performance between male and female students.

7.4. Impact of Study Time

```
study_scores = df.groupby("studytime")[['G1','G2','G3']].mean()
study_scores.plot(kind="bar")
```

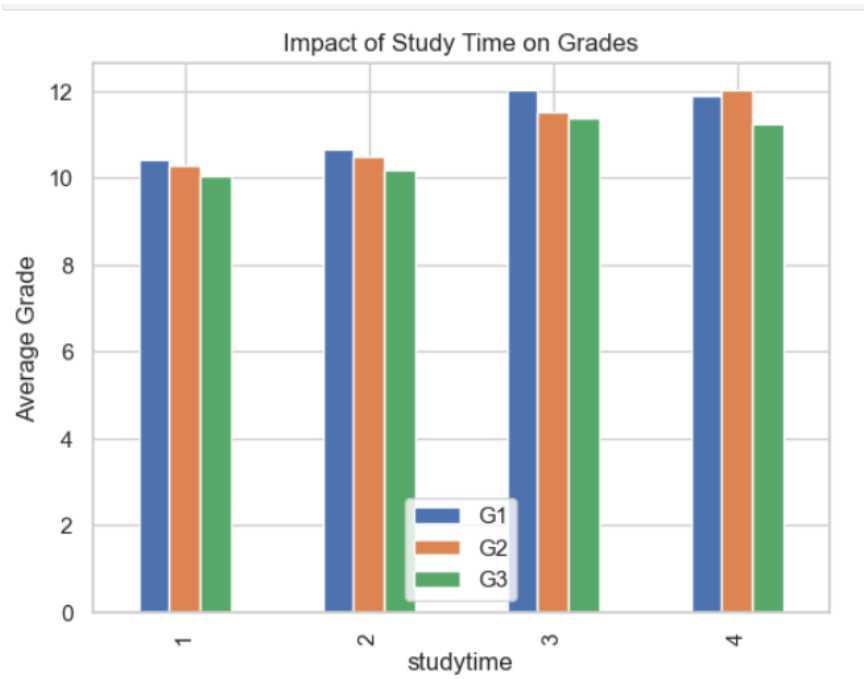

Figure 4: Impact of Study Time on Grades

Interpretation:
Students who study more tend to achieve better grades, showing a positive relationship between study time and performance.

7.5. Correlation Analysis

```
correlation = df[['G1','G2','G3','studytime','absences']].corr()
sns.heatmap(correlation, annot=True, cmap="coolwarm")
```

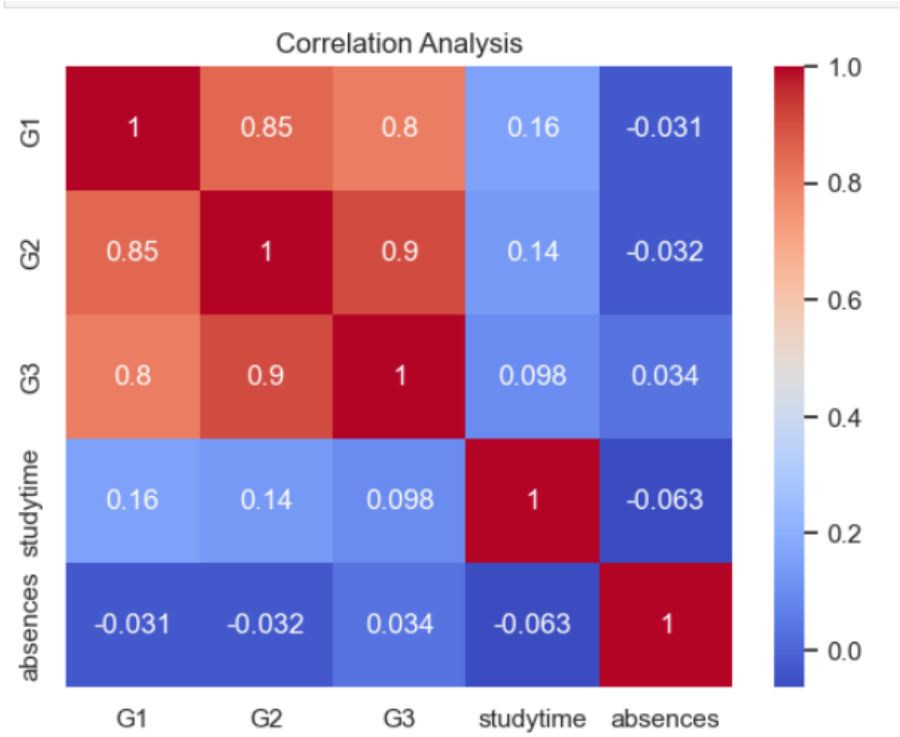

Figure 5: Correlation Heatmap

Interpretation:

The heatmap shows relationships between variables:

- G1, G2, and G3 are highly correlated
- Study time positively affects grades
- Absences negatively affect performance

8. Key Insights

- Students' final grades are strongly related to earlier grades (G1 and G2)
- Higher study time leads to better performance
- Students with more absences tend to score lower
- Pass percentage indicates overall academic success level
- Gender-based differences in performance can be observed

9. Conclusion

This project successfully analyzed student performance using data analysis techniques. The results show that factors like study time and attendance significantly impact student grades.

Such analysis can help educators:

- Improve teaching strategies
- Identify weak students early
- Encourage better study habits

10.Future Improvements

- Use machine learning to predict student performance
- Include more features like parental education and internet access
- Build a dashboard using Power BI or Tableau